

Domain Engineering in Lithium Niobate and Lithium Tantalate: Domain Wall Motion

V. YA. SHUR*

Ferroelectric Laboratory, Ural State University, Ekaterinburg 620083, Russia

The main aspects of domain engineering in LiNbO_3 and LiTaO_3 representing field induced formation of the tailored domains are discussed. The special attention is paid to sideways domain wall motion. Experimentally observed wall motion scenarios have been classified. It is discovered that the realization of the given scenario is defined by degree of deviation of the switching conditions from equilibrium characterized by ratio between switching and screening rates. It has been proved that the domain growth acceleration after merging is due to formation of super-fast X-oriented walls. The loss of wall shape stability and formation of nano-domains have been considered.

Keywords Periodical poling; nano-domains; self-assembling; domain wall; ferroelectric; domain engineering; lithium niobate; lithium tantalate

Introduction

Recent decade the domain engineering is becoming one of the most important areas of ferroelectric science and technology. The main target of the domain engineering is the manufacturing of the stable tailored domain patterns in commercially available ferroelectrics possessing characteristics important for applications. The ability to introduce the spatial modulation of the electro-optic, photorefractive, piezoelectric, and nonlinear optical properties by creation of the periodic domain structure opens the wide possibilities in manufacturing the new devices with record characteristics. The relatively low price of devices can be achieved using lithographic methods for precise periodical poling. The potential for this technology is ready to expand into very large market.

An optimization of the poling process resulting from the fundamental studies is expected to enable the fabrication of sub-micron-pitch gratings and engineered 1D and 2D structures, which could meet the demanding specifications for photonic applications. The ultimate interest is the exploitation of fine (nanoscale) domain structuring capabilities for optical devices.

Let us remind briefly the main stages of the domain engineering history. The idea to optimize the properties of ferroelectrics by creation of the stable tailored domain structures has been advanced thirty years ago by Robert Newnham and Eric Cross with coauthors [1]. They have written about tailored domain patterns in piezoelectric crystals and have discussed the possibility to modify the properties of the crystal by producing “domains which are not to be switched during device operation.” The main discussed area of application

Received November 15, 2005; in final form May 24, 2006.

*Corresponding author. E-mail: vladimir.shur@usu.ru

has been related to piezoelectric devices. The authors believe that the domain structure should be added to the list of important factors which can be manipulated in selecting a desired resonance pattern along with boundary conditions and electrode configurations. “Twinning is generally considered to be a nuisance which detracts from the performance of a piezoelectric oscillator, but this need not be so. If the domain structure is properly designed, the resonant frequency spectrum can be adjusted to enhance or eliminate certain modes. Normally forbidden modes can be generated in this way, and very high frequency modes become possible. There are also applications in the field of acoustics where multi-domain ceramics transducers can be used for focus acoustic power” [1].

It is necessary to mention the parallel development of the independent branch of the domain engineering dealing with creation of the periodically poled nonlinear optical crystals for light frequency conversion (first of all second harmonic generation SHG) and production of quasi-phase-matched non-linear optical devices.

It is known that the phase-matching is the necessary condition for achievement the high light conversion efficiency due to different phase velocities of the interacting light waves. The original method of phase-matching by reversing the sign of the nonlinear coefficient every coherence length (l_c) has been proposed by Armstrong, Bloembergen et al. [2], where l_c is the distance, over which a phase difference over π is accumulated between the interacting waves.

This idea has been applied experimentally to array of ferroelectric domains by R.C. Miller [3]. “The optimum domain array for SHG is one where the crystal consists of sheets of antiparallel domains, each l_c thick, with the wall normals parallel to the beam direction” [3]. He has shown that a SHG intensity enhancement is proportional to $(1 + 2N)$, where N is the number of 180-degree properly placed domain walls in array.

The LiNbO_3 (LN) and LiTaO_3 (LT) single crystals are mostly used for these devices due to their record electro-optic and nonlinear optical properties. Moreover, LN and LT are appropriate for production of the perfect domain arrays, because both crystals are uniaxial with 180-degree domain walls strictly oriented in proper crystallographic direction. The only problem is caused by extremely high values of the threshold fields necessary for poling in congruent crystals (>210 kV/cm for CLN and CLT) hindering the treatment of the wafers with the necessary thickness above 0.5 mm.

Some progress has been achieved on the way to develop the appropriate method for production of periodically poled LN (PPLN) and LT (PPLT). D. Feng and N. Ming [4, 5] were the first, who produced PPLN by growth of doped CLN crystals in periodically variable conditions. As a result, the artificially controlled growth striations lead to formation of the periodic domain structure with periods down to 2 microns. Many other methods have been used for creation of the periodical domain structure including diffusion, proton exchange, electron beam, and others. But all of them do not satisfy the requirements of industrial technology.

The most important step in development of the domain engineering in LN and LT occurred in 1993, when the poling of CLN by application of the external electric field using the electrode pattern produced by photolithography was done by M. Yamada et al. [6]. This result opened the way to mass production of quasi-phase-matched devices. The revolutionary character of realized technique has been pointed out by R. Byer [7]: “The leverage of mass production made possible by lithographic patterning with subsequent domain inversion has led to a rapid transition from nonlinear crystals that cost thousands of dollars to fabricate to nonlinear chips that cost less than one dollar each to fabricate. For the first time, more than thirty years after the first suggestion of quasi-phase-matching by Bloembergen, nonlinear chips can be engineered and mass produced at a cost that is

acceptable for widespread consumer applications.” The further activities have been directed to improvement of the electrical poling method, decreasing of the threshold fields, and struggle with the optical damage.

It is clear that development of the domain engineering requires the deep understanding of the main physical mechanisms governing the formation of the domain structure. Our experimental study of the polarization reversal reveals that the domain structure geometry is of kinetic nature and the real stable domain patterns are always far from equilibrium. The crucial result is that almost any domain pattern can be stabilized by screening of depolarization field [8, 9]. The developed approach based on the main experimental facts allows us to achieve the essential improvement of the poling processes both for micro- and nano-scale domain patterning [8, 9].

The kinetics of the domain structure under application of the spatially nonuniform electric field is a complicated process involving several stages: (1) formation of the new domains (nucleation), (2) domain growth in forward and sideways directions, and (3) coalescence (merging). The domain growth by sideways motion of the domain walls is the key stage for creation of the periodical domain structure by electric poling. It is necessary to point out that while it seems that the motion of the single domain wall is the trivial process well known for decades, it is not the case. The recent studies using modern methods of domain observation with nanoscale resolution provide a new insight in the domain kinetics [9]. The new and unexpected scenarios of the discrete domain growth through formation of self-assembled nanodomain chains, arrays, and fingers were discovered.

This paper represents the review of our most important recent achievements in experimental and theoretical study of the domain wall kinetics in LN and LT single crystals in wide range of experimental conditions. The micro- to nano-scale domain kinetics is discussed. The special attention is paid to formation of self-assembled domain structures.

Domain Wall Motion: Classical Approach

The classical approach to the domain wall motion under the action of electric field is based on the nucleation theory. The wall motion is considered similar to propagation of the phase boundary during any first order phase transition with electric field being the driving force [8, 9]. In this case the layer-by-layer mechanism is typical for “equilibrium growth” of new phase. Two types of nucleation are considered: (1) generation of the elementary steps with thickness about one unit cell at the existing domain wall by 2D nucleation and (2) step growth along the wall in definite crystallographic directions by 1D nucleation (Fig. 1). In equilibrium conditions the probability of 1D nucleation is essentially higher than 2D

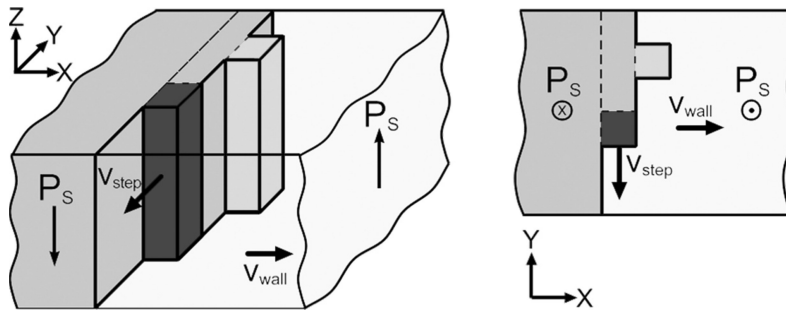


Figure 1. Scheme of sideways domain wall motion by generation and growth of elementary steps.

nucleation and the moving domain walls are almost flat. The plane walls are strictly oriented along proper crystallographic directions.

The situation changes drastically for nonequilibrium conditions, when formation of the spatially nonuniform electric field during switching leads to realization of nontrivial variants of the wall motion. It is important to mention that the local value of switching field (E_s) averaged over the nuclei volume, which drives the nucleation process, is a superposition of external, depolarization, and screening fields [8, 9].

The main problem presents the existence of depolarization field E_{dep} , which always essentially exceeds the experimentally used applied fields. This field is produced by bound charges and is compensated partially by various screening processes. The pronounced effect of E_{dep} is observed during backswitching, which represents spontaneous decay of the domain state produced during poling with ineffective screening after removing of external electric field. The tailored domain pattern decays in this case. Thus, effective screening of E_{dep} is necessary for stabilization of the poled domain structure.

All screening effects can be divided in two groups: (1) external and (2) bulk. In ferroelectric capacitor (plate cut normal to polar axis and covered by the electrodes) the external screening occurs by redistribution of the charges in the electrodes. This fast process without any memory effect plays the main role in E_{dep} compensation. The spatial distribution of screening charges follows the spatial distribution of the bound charges. The retardation of external screening is negligible on usual time scale of the polarization reversal. Nevertheless, the existence of surface intrinsic (or artificial) dielectric/dead layer does not allow to compensate E_{dep} completely [8, 9]. The residual depolarization field E_{rd} being several orders of magnitude smaller than E_{dep} nevertheless is of the order of the threshold (“coercive”) field necessary to apply for starting the polarization reversal.

Compensation of E_{rd} can be realized through redistribution of bulk charges or reorientation of defect dipoles [10, 11]. The carriers injected from electrodes due to extremely high field in the surface dielectric layer can be involved in bulk screening also [12]. All bulk screening processes are comparatively slow. During polarization reversal the time lag of the bulk screening charge redistribution (retardation effect) as compared to the instantaneous bound charge redistribution (corresponding to instantaneous domain pattern) always exists. Several evidences of the retardation effect are observed experimentally: (1) deceleration of the moving wall caused by the following trail of uncompensated bound charge, (2) unipolarity of the polarization reversal, which manifests itself as existence of the bias field characterized by the shift of the hysteresis loop during cyclic switching, (3) reconstruction of the initial domain pattern after removing of external field (“short memory effect”), (4) dependence of the domain structure kinetics on prehistory (“long memory effect”) [8, 9, 13].

The systematic investigation of the domain kinetics in LN and LT single crystals reveals a great variety of the kinetics scenarios leading to formation of qualitatively different domain patterns. The main problem in their classification lies in the fact that it is impossible to attribute the given scenario to a definite range of applied fields and switching rates. The critical values of the fields and rates corresponding to a replacement of one domain kinetics scenario by another differ drastically for different experimental conditions. We have proposed and confirmed both experimentally and by computer simulation that the ratio (R) between switching rate ($1/t_s$) and bulk screening rate ($1/\tau_{scr}$) can be used as the crucial parameter determining the selection of the particular scenario of the domain kinetics including the exotic ones [9]. Thus, the poling with proper value of R allows to control the geometry and parameters of the tailored domain patterns and even to realize the formation of self-assembled nanoscale structures.

Loss of the Domain Wall Shape Stability

“Finger” Structures

In the real experimental situations the essential distortion of the regular domain patterns is observed during polarization reversal. This irregular behavior can be well understood within the framework of above-discussed approach. E_s varies both spatially and temporarily, thus altering the ratio of the probabilities of elementary nucleation processes. The field variation can be induced by: (1) the fringe effect existing for any finite electrode, (2) screening degree of instantaneous domain pattern. The latter effect is the most pronounced, as it was discussed above, when screening of depolarization field is incomplete ($R > 1$) or ineffective ($R \gg 1$) and the classical wall motion must be suppressed.

It has been shown experimentally that under such conditions any random perturbation of the planar wall shape is amplified and as a result of its propagation the self-assembled structure appears. We observed experimentally the peculiarities of polarization reversal in CLN in the area covered by artificial dielectric layer, where the condition $R > 1$ is fulfilled. As it is seen, the loss of domain wall shape stability results in formation of a ledge at the domain wall (“finger”). The sub-micron finger structure develops by propagation along the wall through generation of neighboring fingers and their simultaneous extension along one and the same direction (Fig. 2). It must be emphasized that emerging fingers are parallel and strictly oriented along Y crystallographic direction.

All main features of such loss of the domain wall shape stability: (1) quasi-periodic discrete character of finger structure, (2) its self-maintain propagation, and (3) strict orientation are well understood, if we take into account the decisive role of E_{rd} . The unscreened field of bound charges of freshly switched finger suppressing 2D nucleation prevents its lateral expansion restricting the transverse size of the finger. The electrostatic repulsion induced by the same charges leads to arising of new domains at the proper distance from each other (correlated nucleation) [9, 14]. The subsequent growth of the front of the finger structure is caused by the field maximum arising at the tips of the fingers. The E_{rd} accelerates 2D nucleation at the tips of the fingers conserving the orientation of the front of the finger structure. The individual finger orientation is also caused by E_{rd} . It has been demonstrated experimentally that the domain growth under highly non-equilibrium conditions is essentially anisotropic [9].

During periodical poling the described mechanism of domain motion provides the large shift of the domain wall from the electrode edge (abnormal “domain broadening”). This extremely undesirable effect leads to merging of the neighboring domains, thus destroying the domain pattern periodicity [15–17].

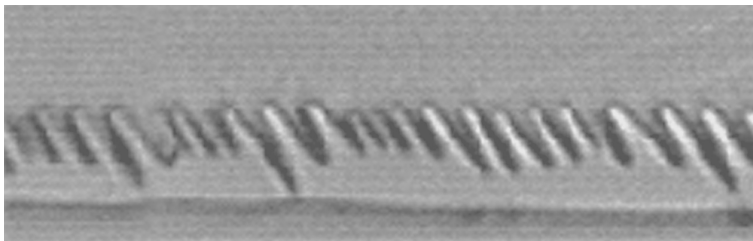


Figure 2. The finger structure formed during wall motion in CLN. Optical microscopy. Domain pattern revealed by etching.

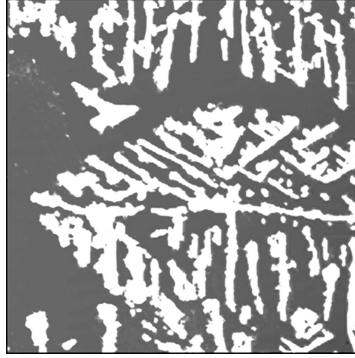


Figure 3. Dendrite domain structure. AFM image. Domain pattern revealed by etching.

Dendrite Structures

The unusual domain patterns are formed during spontaneous backswitching after removing of electric field. The backswitching poling procedure proposed by us and described in detail in Ref. [9, 18, 19] allows us to create the dendrite domain structure (Fig. 3), which covers large areas (about several square millimeters). Its main feature is that the orientation along the crystallographic directions is lost for the domain ensemble as a whole, but the local symmetry still holds. The loss of long-range order can be attributed to the prevalence of super-strong backswitching field (E_{rd}) over the long-range electrostatic interaction of bound charges.

In some experimental conditions the enlargement of the dendrite structure is achieved through propagation of strictly oriented plane front (Fig. 3). As it is seen, the growing arrays of fingers choose one of the three equivalent Y crystallographic directions. It must be mentioned that the domains comprising the observed structures are non-through and thus possess charged domain walls.

Domain Growth Acceleration Induced by Merging

It has been shown by direct observation that the domain growth essentially accelerates after domain merging. The formation of large polygon domain after merging of two individual small polygons is so fast that it is impossible to observe the details of the shape evolution. The effective growth velocity increases by several orders of magnitude. One of the interesting examples of the domain growth acceleration is observed for domain wall interacting with the defect (Fig. 4). The coexistence on the optical images of both past and present positions of the domain walls is the bright demonstration of jump-like domain growth (Fig. 4b).

The detail experimental study and computer simulation reveal that the super-fast domain walls oriented along X direction are formed after merging. These super mobile X walls appear, grow, and disappear during the short time interval (“life time”). Their record high mobility is due to the fact that during the whole life time these walls are characterized by the maximum possible concentration of the elementary steps. The crucial role of this super-fast switching process is obtained for high concentration of the individual domains, which is typical for CLT [20, 21].

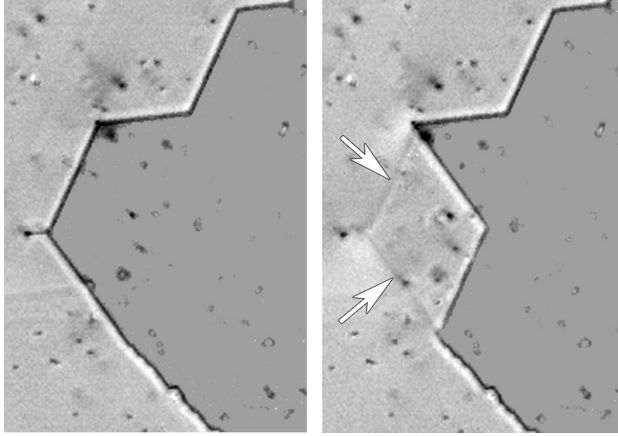


Figure 4. Jump-like domain growth through formation of super-fast X domain wall in CLN. Time interval between images 40 ms. In-situ optical observation.

Super-Fast Domain Growth

The most strange domain kinetics is obtained, when the bulk screening becomes absolutely ineffective ($R \gg 1$). In this case both above discussed alternative mechanisms of the domain wall motion are forbidden. Nevertheless, the experiment shows that the switching process takes place even under such unfavorable conditions, demonstrating qualitatively new scenarios of domain structure evolution and exotic discrete mechanisms of domain growth.

Two types of experiments revealing the key role of the intrinsic or artificial surface dielectric layer were carried out: (1) the switching in ferroelectric capacitor with artificial dielectric surface layers of given thickness and (2) spontaneous backswitching induced by fast removing of the external field. Several types of discrete structures have been produced experimentally: (1) long individual domain rays strictly oriented along crystallographic directions (Fig. 5), (2) self-organized ensembles of micro- and nano-scale domain arrays (Fig. 6). The arising and formation of the later structures were observed during “super-fast switching” in lead germanate PGO [22, 23], LN and LT single crystals.

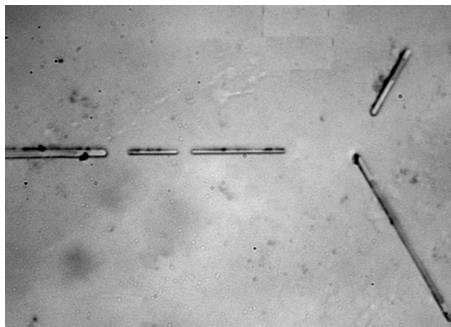


Figure 5. Growth of domain rays oriented along Y direction. Switching under uniform electrode in CLN. Optical observation of the domains revealed by etching.

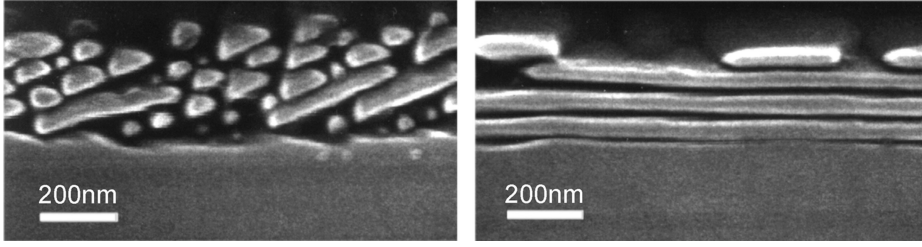


Figure 6. SEM images of (a) nano-domain arrays and (b) quasi-periodic stripe nano-domain structure. Domain patterns are formed as a result of backswitching near the edge of the stripe electrode under the artificial dielectric layer in CLN. Domains revealed by etching.

Switching with Artificial Surface Dielectric Layer. We have switched 2-mm-thick stoichiometric LT (SLT) sample covered by artificial dielectric layer (2- μm -thick photoresist) using liquid electrodes. The switching process starts from the formation of a small hexagon domain around the pin-hole in the dielectric layer. The subsequent growth is achieved through propagation of the quasi-regular ensemble of submicro-scale isolated domains (Fig. 8). The spreading of this structure follows the rules proposed by the nucleation theory for the explanation of the growth of the continuous isolated domain [9, 20]. The only difference is that in this case the boundary of the area covered by isolated switched domains plays the role of the domain wall. It is seen that the propagation of the boundary is limited by “step generation” and “step growth” (Fig. 7). Moreover, similar to the observed usual growth of macro-domain, the discretely switched area represents a regular polygon oriented along crystallographic directions.

The statistical analysis of the experimental data shows that the averaged distance between the neighboring isolated micro-domains comprising the structure is close to the thickness of the artificial dielectric layer. It is in accord with the discussed crucial role of E_{rd} and its dependence on the gap thickness. It must be stressed that revealed correlation

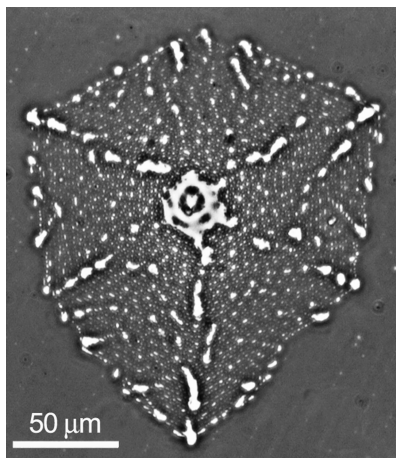


Figure 7. Formation of self-assembled domain structure around the pinhole in the dielectric layer in 2 mm-thick SLT with Z+ surface covered by 2 μm -thick photoresist. Optically visualized domain structure revealed by etching.

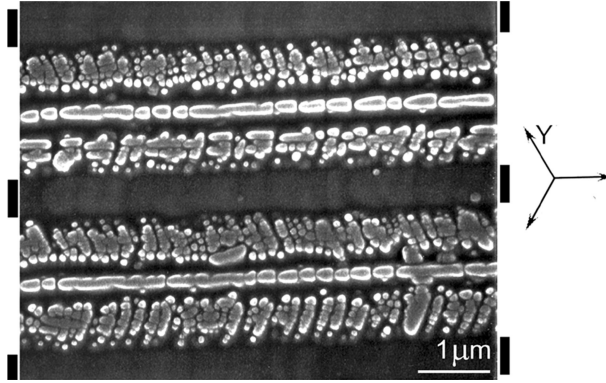


Figure 8. Nanodomain arrays oriented along Y directions appeared during backswitched periodical poling of CLN. Domain patterns revealed by etching and visualized by SEM. Black rectangles – electrode position.

is important for evaluation of the effective depth of the intrinsic dielectric gap in proper material and experimental condition by measuring the average period of the quasi-regular structure formed during discrete switching.

Nanoscale Domain Arrays. The formation of nano-scale self-assembled structures during backswitching was studied in CLN. The backswitching is very fast in CLN, because it occurs under the action of extremely high fields due to record-breaking value of spontaneous polarization leading to abnormally high value of the bulk screening field. After external field switch-off the local field changes the sign for incomplete screening leading to a spontaneous backward switching. It has been shown experimentally that the backward motion of the domain wall is achieved through propagation of the boundary of the ensemble of isolated nano-scale domains (Fig. 8). It is important to point out that under these experimental conditions the forward and backward wall motions differ qualitatively. The forward motion occurs, when the applied field slightly exceeds the abnormally high bulk screening field. Thus, even when the applied field value exceeds the coercive value ($E_c = 210 \text{ kV/cm}$), the switching field is comparatively low. In this case the classical low field wall motion scenario is realized. Contrary, the backward motion is governed by the abnormally high bulk screening field itself and its relaxation is absolutely ineffective. This situation can be attributed to the case $R \gg 1$.

The formed domain patterns revealed by etching and visualized by SEM (Fig. 8) demonstrate the reversal motion of the existing domain walls through propagation of the highly organized quasi-periodical structure of domain arrays oriented along crystallographic directions [16, 18, 19, 24]. Each quasi-regular array consists of nano-domains with a diameter 30-100 nm and an average linear density exceeding 10^4 mm^{-1} . Such domain evolution scenario was observed during switching using periodical stripe electrodes oriented along the Y-direction deposited on Z^+ -surface.

All effects observed during backswitching can be explained by the nonuniform distribution of the local backswitching field near the electrode edges and domain walls due to spatially nonuniform screening of the external and depolarization fields. Our estimations reveal the field maximum at a distance of about the thickness of the insulating layer from the domain wall or nano-domain array. An array aggregate existing at a given moment

generates the new maximum of the backswitching field at a fixed distance in front of its boundary and triggers the formation of the new array. The developing correlated nucleation leads to the self-maintaining generation of the parallel arrays.

Theoretical Background: Effect of Correlated Nucleation

The observed self-assembled behavior of arising domain patterns can be well understood within the framework of considered nucleation theory of domain growth, if the decisive role of uncompensated residual depolarization field in formation of local switching field is accounted for. The trail of uncompensated depolarization field exists behind the fast-moving domain wall due to artificially induced or inherent ineffectiveness of bulk screening. The numerical solution of 2D electrostatic problem in ferroelectric capacitor reveals that the instantaneous value of E_s demonstrates a pronounced maximum at some distance from the moving domain wall (Fig. 9). While the field maximum diminishes with the depth, nevertheless, this field singularity close to the surface drastically changes the scenario of the domain kinetics. The existence of the field maximum explains why the nucleation at some distance from the wall prevails. The computer simulations show that this distance is nearly equal to the thickness of the surface dielectric layer L . It is easy to generalize this result for the explanation of above discussed experiments with artificial dielectric gap. It is important to take in mind that the propagating “boundary” of the enlarging discrete domain ensemble also produces the same field maximum as in the case of “continuous” domain.

On one hand, it can be shown that any isolated domain arisen within this maximum cannot spread out due to the same effect of the uncompensated depolarization field. On the other hand, the effect of long-range repulsion of the arising domains can not be ignored. This effect can be accounted for as the decrease of the local electric field in the vicinity of each isolated domain thus resulting in formation of isolated domains at definite distance from each other. The experiment shows that the quasi-regular domain chain aligned along the wall consists of needle-like domains with submicro- or nano-scale lateral sizes [9].

It has been shown by simulation that the new field maximum appears at approximately the same distance from the formed domain chain, thus initiating formation of the new chain. This process can reproduce itself infinitely. It has been shown experimentally that such fast growing self-assembled structures can cover areas of about square millimeters.

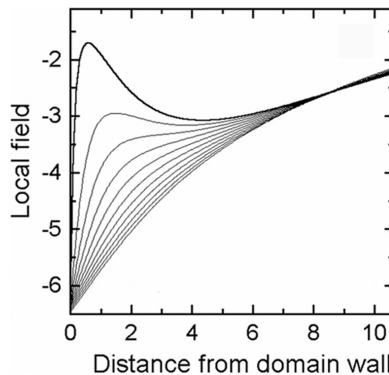


Figure 9. The simulated local field distribution near the stripe domain at different depth from the dielectric surface layer ranged from $0.1 L$ to $3.9 L$. The distance from the wall is divided by the thickness of the surface dielectric layer L .

All above considerations make it possible to assign all observed experimental peculiarities to the manifestation of one and the same self-maintained process, which we will name “correlated nucleation.” It must be stressed that this effect can be observed only if the switching process takes place under highly non-equilibrium conditions when $R \gg 1$. That is why it can not be understood and explained within the traditional thermodynamic approach. We have shown that the proper explanation requires the development of adequate kinetic theory. Moreover, the strong deviations from thermodynamically equilibrium state explain the occurrence of the preferred directions of kinetic “fluxes” (the “finger orientation”)—the effect, which is well-known in the other fields of modern physics [25].

There are two ways to intensify the role of the correlated nucleation process. One of them has been discussed above and is realized by increasing of the ratio R , by amplification of the switching field, e.g., by making use of edge effect. The alternative possibility is the deposition of the artificial surface dielectric layer increasing the residual depolarization field. In this case, the period of the observed quasi-regular structure increases proportionally to the thickness of the layer.

Application for Periodical Poling

Finally, it will be shown, how the above discussed knowledge of the domain structure evolution has been used for production of PPLN and PPMgO:LN quasi-phase-matched devices for SHG with high efficiency.

The commonly used electric field poling technology, which is based on the application of a high voltage pulse or number of short pulses through lithographically defined electrodes deposited on Z^+ surface, cannot be straightforwardly applied to the fabrication of sub-micron gratings. Thus, for nano-scale domain engineering such alternative technique as “self-organized domain patterning” due to self-maintained evolution of domain structure is under consideration [18, 19].

The waveform for backswitched poling consists of three levels of external field: “high field”, “low field”, and “stabilization field” (Fig. 10). Such waveform allows one to control the different stages of domain evolution during poling. Moreover, we exploit the unique abilities of the backswitching process, which was always considered as undesirable one, because it destroyed the tailored structure.

Several distinguishable stages of domain evolution can be revealed during poling (Fig. 11). The process starts with formation of new domains at the Z^+ -polar surface along the electrode edges. The position of the nuclei arrays is due to the field singularities caused

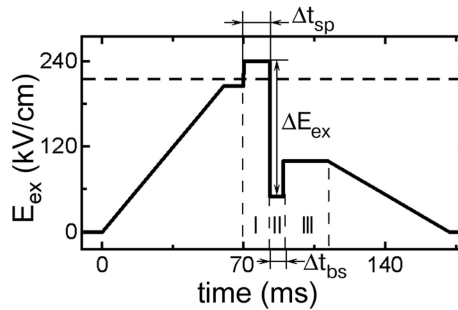


Figure 10. Voltage wave-form for backswitch poling. Dash line corresponds to threshold value of external field.

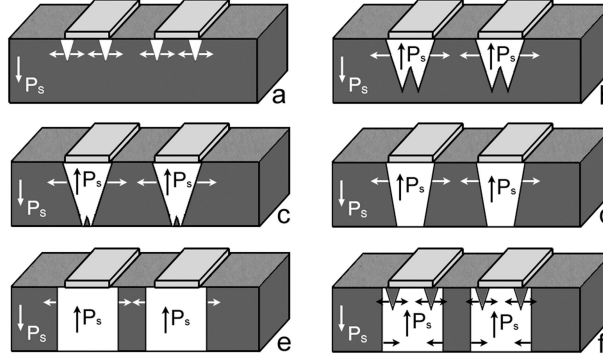


Figure 11. Main stages of domain evolution during periodical poling.

by the fringe effect (Fig. 11a). The real value of E_s along the electrode edge alters profoundly with small deviations of its shape, which can introduce the undesirable field fluctuations. During the second stage, the formed domains grow and propagate through the wafer. As a result of merging, laminar domains with plane walls are formed. The main problem of the periodical poling is the essential domain broadening out of the electroded area, which is always observed (Fig. 11b). This effect leads to an essential difference between the lithographically defined electrode pattern and the produced periodical domain structure. For short periods domain broadening results in domain merging, thus limiting the production of the short-pitch domain patterns. During broadening the domain wall moves in the regions with the artificial surface dielectric layer; thus, above discussed exotic scenarios of the domain wall motion can be realized. The formation of the domain fingers and arrays leads to abnormally high domain broadening, which is extremely undesirable for periodical poling.

After rapid decreasing of the poling field, the backswitching starts through shrinkage of the laminar domains by the backward wall motion and nucleation of the domains with the initial orientation of spontaneous polarization along the electrode edges (Fig. 11c). As it was discussed above, the backswitching occurs under the action of the abnormally high field. That is the reason why various regular self-assembled nano-domain structures

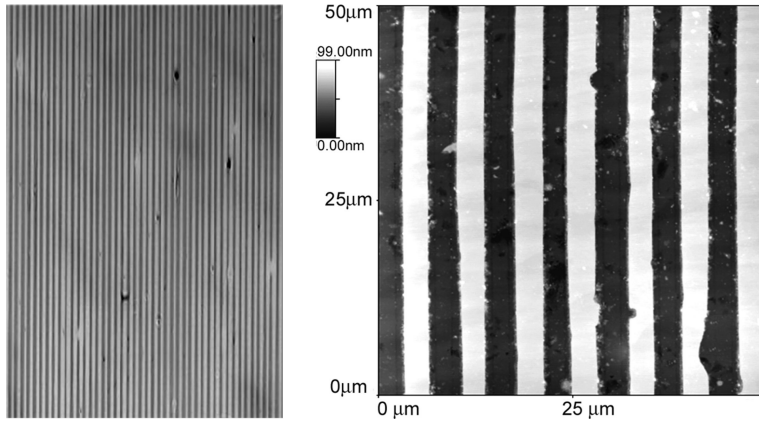


Figure 12. Periodical tailored domain structure in MgO:LN.

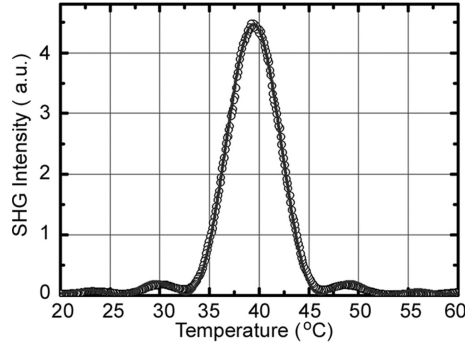


Figure 13. Second harmonic generation (1064 nm $\rightarrow$ 532 nm) temperature tuning curve in MgO:LN with periodical domain structure 1 mm thick, 5 mm length, 6.95 μm grating.

can be produced by backswitched poling by presetting the voltage wave-form parameters. The types of domain patterns depend also on the value of the domain wall shift out of the electrodes. The switching from the single-domain state takes place at the “high field,” and the backswitching occurs at the “low field.” The crucial parameters for backswitching kinetics are the duration of the “high field” stage Δt_{sp} and the field-diminishing amplitude ΔE (Fig. 10). The application of this improved poling method to LN and LT single crystals demonstrates the spatial frequency multiplication of the domain patterns as compared to the spatial frequency of the electrodes [9] and self-maintained formation of the oriented domain arrays consisting of individual nano-scale domains with diameter down to 30 nm and average linear density exceeding 10^4 mm^{-1} (Fig. 8) [18, 19].

The obtained knowledge of the domain wall motion growth allows us to optimize the periodical poling technology for production of the precise periodic domain structures with periods down to 4 μm in 1-mm-thick wafers of MgO:LN (Fig. 12).

The quality of the tailored domain structures for nonlinear-optical application can be tested by measurements of the SHG temperature tuning curve (Fig. 13). The coincidence between theoretical prediction for ideal periodical domain structure and experimental data demonstrates the high quality of the periodical pattern.

Conclusion

The variety of experimentally observed scenarios of the sideways domain wall motion in LN has been classified. It has been demonstrated that the value of the ratio between switching and screening rates is the critical parameter defining the realized scenario. The observed unusual domain wall motion has been attributed to deceleration of the wall motion by classical layer-by-layer growth mechanism induced by the trail of uncompensated depolarization field which followed the moving domain wall. The drastic acceleration of the wall motion after merging of neighboring domains has been experimentally observed and attributed to formation of super-fast X-oriented walls with maximal concentration of the elementary steps.

Acknowledgments

The research was made possible in part by Grant 03-51-6562 of INTAS; by Grant 04-02-16770 of RFBR; by Grant 03-02-39004 of RFBR-NNSF; by Grant 04-02-04007 of RFBR-DFG and by Grants 48859, 49130 and RNP 2.1.1.8272 of Program “Development

of the Scientific Potential of High Education” of Federal Agency of Education; by Award No. EK-005-X1 of the U.S. CRDF BRHE and Federal Agency of Education; by Grant RUE1-5037-EK-04 of CRDF FSTM.

References

1. R. E. Newnham, C. S. Miller, L. E. Cross, and T. W. Cline, *Phys. Stat. Sol. (a)* **32**, 69 (1975).
2. J. A. Armstrong, N. Bloembergen, J. Ducuing, and P. S. Perlsan, *Phys. Rev.* **127**, 1918 (1962).
3. R. C. Miller, *Phys. Rev.* **134**, A1313 (1964).
4. D. Feng, N. B. Ming, J. F Hong, Y. S. Yang, J. S. Zhu, Z. Yang, and Y. N. Wang, *Appl. Phys. Lett.* **37**, 607 (1980).
5. D. Feng, N. B. Ming, and D. Feng, *J. Mater. Sci.* **17**, 1663 (1982).
6. M. Yamada, N. Nada, M. Saitoh, and K. Watanabe, *Appl. Phys. Lett.* **62**, 435 (1993).
7. R. L. Byer, *J. Nonlinear Opt. Phys. & Mater.* **6**, 549 (1997).
8. V. Ya. Shur, in “*Ferroelectric Thin Films: Synthesis and Basic Properties*,” 153 (ed. by C. A. Paz de Araujo, J. F. Scott, and G. W. Taylor, Gordon and Breach, New York, 1996).
9. V. Ya. Shur, in “*Nucleation Theory and Applications*” 178 (Ed. by J. W. P. Schmelzer, WILEY-VCH, Weinheim, 2005).
10. V. M. Fridkin, “*Ferroelectrics Semiconductors*” (Consult. Bureau, New York and London, 1980).
11. P. V. Lambeck and G. H. Jonker, *J. Phys. Chem. Solids* **47**, 453 (1986).
12. A. K. Tagantsev, I. Stolichnov, E. L. Colla, and N. Setter, *J. Appl. Phys.* **90**, 1387 (2001).
13. V. Ya. Shur, *Journal of Materials Science* **41**, 199 (2006).
14. V. Ya. Shur, E. L. Rumyantsev, E. V. Nikolaeva, and E. I. Shishkin, *Integrated Ferroelectrics* **59**, 1493 (2003).
15. G. Rosenman, A. Skliar, and A. Arie, *Ferroelectrics Review* **1**, 263 (1999).
16. V. Ya. Shur, E. Rumyantsev, E. Nikolaeva, E. Shishkin, R. G. Batchko, G. D. Miller, M. M. Fejer, and R. L. Byer, *SPIE Proc. on Smart Structures and Materials* **3992**, 143 (2000).
17. V. Shur, E. Rumyantsev, R. Batchko, G. Miller, M. Fejer, and R. Byer, *Ferroelectrics* **221**, 157 (1999).
18. V. Ya. Shur, E. L. Rumyantsev, E. V. Nikolaeva, E. I. Shishkin, D. V. Fursov, R. G. Batchko, L. A. Eyres, M. M. Fejer, and R. L. Byer, *Appl. Phys. Lett.* **76**, 143 (2000).
19. V. Ya. Shur, E. L. Rumyantsev, E. V. Nikolaeva, E. I. Shishkin, D. V. Fursov, R. G. Batchko, L. A. Eyres, M. M. Fejer, R. L. Byer, and J. Sindel, *Ferroelectrics* **253**, 105 (2001).
20. V. Ya. Shur, E. V. Nikolaeva, E. I. Shishkin, A. P. Chernykh, K. Terabe, K. Kitamura, H. Ito, and K. Nakamura, *Ferroelectrics* **269**, 195 (2002).
21. V. Ya. Shur, E. V. Nikolaeva, E. I. Shishkin, V. L. Kozhevnikov, A. P. Chernykh, K. Terabe, and K. Kitamura, *Appl. Phys. Lett.* **79**, 3146 (2001).
22. V. Ya. Shur, A. L. Gruverman, N. Yu. Ponomarev, and N. A. Tonkachyova, *Ferroelectrics* **126**, 371 (1992).
23. V. Ya. Shur, A. L. Gruverman, N. Yu. Ponomarev, E. L. Rumyantsev, and N. A. Tonkachyova, *Integrated Ferroelectrics* **2**, 51 (1992).
24. R. G. Batchko, V. Y. Shur, M. M. Fejer, and R. L. Byer, *Appl. Phys. Lett.* **75**, 1673 (1999).
25. H. Haken, *Synergetics* (Berlin/N.Y.: Springer, 1978).